

1. Show that for a regular polygon with n sides ($n > 5$), the number of triangles whose vertices are non-adjacent vertices of the polygon is $n(n-4)(n-5)/6$.
2. Each face of a cube is coloured by a different color. How many of the colourings are distinct.
3. Let A be a $2n$ -element set where $n > 1$. Find the number of different pairings of A .
4. For a positive integer n , the n -th subdivision of an equilateral triangle ABC is the configuration obtained by (i) dividing each side of $\triangle ABC$ into $n+1$ equal parts by n points; and (ii) adding $3n$ line segments to join the $3n$ pairs of such points on adjacent sides so that the line segments are parallel to the third side. Find the number of parallelograms in the n -th subdivision of an equilateral triangle.
5. Let $A \subseteq \{1, 2, \dots, 2n\}$ such that $|A| = n+1$. Show that there exist $a, b \in A$, with $a \neq b$ such that $a|b$.
6. Let $X \subset \{1, 2, \dots, 99\}$ and $|X| = 10$. Show that it is possible to select two disjoint non-empty proper subsets Y, Z of X such that sum of the elements in the set Y is equal to the sum of the elements of the set Z .
7. Let $A = \{a_1, a_2, \dots, a_{2k+1}\}$, where $k \geq 1$, be a set of $2k+1$ positive integers. Show that for any permutation $a_{\pi(1)}, a_{\pi(2)}, \dots, a_{\pi(2k+1)}$ of A , the product $\prod_{j=1}^{2k+1} (a_{\pi(j)} - a_j)$ is always even.
8. Prove that the decimal expansion of a rational number is eventually repeating. What about the converse?
9. Consider a set of $2n$ points in space, $n > 1$. No three are collinear. Prove that maximum n^2 segments (with end points from the given points) can be drawn without forming any triangle with vertices from the given points.
10. Suppose that to every point of the plane a color, either red or blue, is associated. Show that there must exist an equilateral triangle with all vertices of the same colour.
11. Forty-one rooks are placed on a 10×10 chessboard. Prove that you can choose five of them that do not attack each other. (We say that two rooks "attack" each other if they are in the same row or column of the chessboard.)
12. There is a $2n \times 2n$ array (matrix) consisting of 0s and 1s and there are exactly $3n$ zeros. Show that it is possible to remove all the zeros by deleting some n rows and some n columns. Also show that, given any positive integer n there exists a arrangement of $3n+1$ zeroes such that the 0's can not be removed by deleting any n rows and n columns.
13. If every point of the plane is painted one of three colors, do there necessarily exist two points of the same color exactly one inch apart?
14. Let $a_1 < a_2 < \dots < a_{43} < a_{44}$ be positive integers not exceeding 125. Prove that among the 43 consecutive differences $d_i = a_{i+1} - a_i$, some value must occur at least 10 times.
15. How many subsets of $\{1, 2, \dots, n\}$ contain no two consecutive integers.

16. In a given circle, $n \geq 2$ arbitrary chords are drawn such that no three are concurrent within the interior of the circle. Suppose m is the number of points of intersection of the chords within the interior. Find, in terms of n and m , the number r of line segments obtained through dividing the chords by their points of intersection.

17. Suppose that m chords of a given circle are drawn in such a way that no three are concurrent in the interior of the circle. If n denotes the number of points of intersection of the chords within the circle, show that the number of regions divided by the chords in the circle is $m + n + 1$.

18. For $n \geq 4$, let $r(n)$ denote the number of interior regions of a convex n -gon divided by all its diagonals if no three diagonals are concurrent within the n -gon. For instance, $r(4) = 4$ and $r(5) = 11$. Prove that $r(n) = \binom{n}{4} + \binom{n-1}{2}$.

19. There are $p \geq 6$ points given on the circumference of a circle, and every two of the points are joined by a chord. Assume that no 3 chords are concurrent within the interior of the circle

- (i) Find the number of such chords.
- (ii) Find the number of points of intersection of these chords within the interior of the circle.
- (iii) Find the number of line segments obtained through dividing the chords by their points of intersection.
- (iv) Find the number of triangles whose vertices are the points of intersection of the chords within the interior of the circle.

20. A group of n people are standing in a circle, numbered consecutively clockwise from 1 to n . Starting with person no. 2, we remove every other person, proceeding clockwise. For example, if $n = 6$, the people are removed in the order 2, 4, 6, 3, 1, and the last person remaining is no. 5. Let $j(n)$ denote the last person remaining. Find some simple way to compute $j(n)$ for any positive integer $n > 1$.

21. Let a_n denote the number of ways to express n as a sum of positive integers greater than 1, taking *order into account*. For example $a_2 = 1$ since $2 = 2$ is the only representation of 2 in the required form. Similarly $a_3 = 1$. $a_4 = 2$ because $4 = 4$ and $4 = 2 + 2$ are two representations of 4 in required form. $a_5 = 3$ because $5 = 5$, $5 = 2 + 3$ and $5 = 3 + 2$. Find a recurrence relation for (a_n) . Is (a_n) an well known sequence?

22. Prove using a combinatorial argument

$$\sum_{k=1}^n \sum_{i=0}^{k-1} \binom{k-1}{i} \binom{n-k}{i} = 2^{n-1}.$$

23. Prove using a combinatorial argument

$$\sum_{r=0}^n r \binom{n}{r}^2 = n \binom{2n-1}{n-1}.$$

24. Show that

$$\sum_{r=1}^n \frac{(-1)^{r-1}}{r} \binom{n}{r} = \sum_{k=1}^n \frac{1}{k}.$$

25. Consider all r -element subsets of the set $\{1, 2, \dots, n\}$. Each of these subsets has a smallest member. Prove that arithmetic mean of these smallest numbers is $(n+1)/(r+1)$.

26. Prove that

$$\binom{2n}{0}^2 - \binom{2n}{1}^2 + \binom{2n}{2}^2 - \dots - \binom{2n}{2n-1}^2 + \binom{2n}{2n}^2 = (-1)^n \binom{2n}{n}.$$

27. Find the gcd of the following numbers

$$\binom{n}{r}, \binom{n+1}{r}, \binom{n+2}{r}, \dots, \binom{n+r}{r}.$$

28. Find the gcd of the following numbers

$$\binom{2n}{1}, \binom{2n}{3}, \dots, \binom{2n}{2n-1}.$$

29. Let n be a positive integer. Determine a general formula for $\phi(n)$.

30. Suppose there are n letters $L_1, \dots, L_n$ and n envelopes $E_1, \dots, E_n$. Determine number of ways to put the letters in the envelopes such that L_i does not go to E_i for all $1 \leq i \leq n$.

31. Let A and B be two finite sets with $|A| = n$ and $|B| = m$. Find number of onto functions $f : A \rightarrow B$.

32. Consider a $l \times b$ rectangle divided in lb unit squares in the usual way. A diagonal is drawn. How many unit squares will contain a positive length of the diagonal.

33. Each of the vertices of a regular nonagon has been coloured either red or blue. Prove that there exist two congruent monochromatic triangles; that is, triangles whose vertices are all the same color.

34. Finitely many cards are placed in two stacks with more cards in the left stack than the right. Each card has one or more distinct names written on it, although different cards may share some names. For each name we define a shuffle by moving every card that has that name written on it to the opposite stack. Prove that it is always possible to end up with more cards in the right stack by picking several distinct names and doing in turn the shuffle corresponding to that name.

35. The increasing sequence $1, 3, 4, 9, 10, 12, \dots$ consists of all those positive integers that are perfect powers of 3 or sums of distinct powers of 3. Find the 100'th term of this sequence.

36. In how many ways can one arrange 5 indistinguishable armchairs and 5 indistinguishable armless chairs around a circular table? Here two arrangements are considered the same if one can be obtained from the other by rotation.

- 37.** If each row of a rectangular array of distinct real numbers is an increasing sequence, is the same thing true of the rectangular array obtained by rearranging each column to be an increasing sequence (read from top to bottom)?
- 38.** Can a thousand people seated around a circle in seats numbered from 1 to 1000, each person bearing one of the numbers from 1 to 1000, be re-seated so as to preserve their (circular) order and so that no person's number is the same as that of his chair?
- 39.** Let $\mathcal{C}$ be a class of subsets of an n -element set S . Suppose every pair of sets in $\mathcal{C}$ have at least one element in common. Show $|\mathcal{C}| \leq 2^{n-1}$ and equality can be attained.
- 40.** At n distinct points of a circular race course, n cars are ready to start. Each of them covers the course in an hour. At a given signal every car selects one of the two possible directions and starts immediately. Whenever two cars meet both of them change directions and go on without loss of speed. Show that at a certain moment each car will be at its starting point.
- 41.** Each of n boys attends a school gathering with both of his parents. In how many ways can the $3n$ people be divided into groups of three such that each group contains a boy, a male parent and a female parent, and no boy is with both of his parents in his group?
- 42.** A man has 6 friends. At dinner in a certain restaurant, he has met each of them 12 times, every two of them 6 times, every three of them 4 times, every four of them 3 times, every five twice and all six only once. He has dined out 8 times without meeting any of them. How many times has he dined out altogether?
- 43.** Three identical black balls, four identical red balls and five identical white balls are to be arranged in a row. Find the number of ways that this can be done if all the balls with the same colour do not form a single block.
- 44.** In how many ways can a $2 \times n$ board can be filled by n many 2×1 dominoes?
- 45.** In how many ways n open parenthesis (and n close parenthesis) can be arranged in a row correctly?
- 46.** In how many ways a regular n -gon can be triangulated by joining vertices by diagonals so that the diagonals do not intersect each other in the interior of the polygon?
- 47.** If $2n$ people are seated around a circular table, in how many ways can all of them be simultaneously shaking hands with another person at the table in such a way that none of the arms cross each other?
- 48.** A rooted binary tree is an arrangement of points (nodes) and lines connecting them where there is a special node (the root) and as you descend from the root, there are either two lines going down or zero. Internal nodes are the ones that connect to two nodes below. Find the number of rooted binary trees with n internal nodes.